Selecting the Right Mode: Symmetry-Aware, Orbit-Consistent Tesseract Trees on Frozen Foundation Features for 6D Object Pose Estimation

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Abstract

Single-view 6D object pose estimation often fails not because the correct pose cannot be represented, but because it is not selected*: symmetric objects have several equally correct rotations, and near-symmetric ones have look-alike rotations that are wrong. We study this selection problem with a deliberately lightweight design. Two frozen foundation backbones (DINOv3 for appearance, MoGe-2 for surface geometry) feed small autoregressive Transformers that predict a rotation as a root-to-leaf path in a hierarchical tesseract partition of $SO(3)$. The two branches are fused per tree level by a training-free product of experts, and a translation block on the same tokens completes the 6D pose. Three components address selection: (i)* symmetry-aware multi-path targets *that spread the training signal over all equivalent rotations; (ii)* orbit voting*, which scores hierarchical hypotheses for consistency across in-plane rotations of the crop through an exactly derived camera-roll transform; and (iii)* orbit distillation *of that consistency into single-view students. We evaluate on T-LESS, SYMSOL-T, YCB-V and the synthetic-to-real benchmark SPEED+, training only 2.4 M parameters per branch. Multi-path targets reduce the SYMSOL-T error from $3.82°$ to $2.07°$. On SPEED+, orbit voting and distillation reduce the real-domain error from $26.9°/36.7°$ to $11.2°/11.1°$. With synthetic training data, YCB-V drops from $81°$ to $10.9°$. Absolute accuracy remains below fine-tuned state-of-the-art methods, and we report matched comparisons, pre-registered ablations and negative results throughout.*

1. Introduction

Estimating the 6D pose of a known object from one image underlies robotic grasping, augmented reality and autonomous rendezvous. A large part of the remaining error comes from *ambiguity*. A symmetric object has several rotations that produce the same image, and every one of them is correct. A nearly symmetric object has look-alike rotations that differ only in small details, and only one of them is correct. Under a domain shift (renders to real images, new lighting), those small details are exactly what becomes unreliable. Distributional pose methods [7, 13, 16] can represent such multi-modal posteriors, but a single prediction still has to *select* a mode.

We study this selection problem with a deliberately lightweight design built on *frozen* foundation models. DINOv3 [22] gives robust appearance features, and MoGe-2 [24] gives per-pixel surface normals. A small head predicts the rotation as a path in a hierarchical partition of $SO(3)$ derived from the recursive tesseract subdivision [15]. It is an autoregressive Transformer that first chooses one of 4 charts, then one of 8 children at each of 5 levels (131,072 cells), attending to the frozen tokens. The DINOv3 and MoGe-2 trees are fused per level by a training-free product of experts (PoE) [9], and a translation block on the same tokens completes the 6D pose. Only 2.4 M parameters per branch are trained.

Hierarchical hypotheses make selection explicit, and we attack it at three points. *During training*, the target of a symmetric object is the whole set of equivalent rotations, so the tree learns to split its probability between them instead of fitting an arbitrary label (Sec. 3.3). *At test time*, we use the fact that rotating the crop in the image plane is an exact camera roll. A wrong look-alike pose tends to win only in some rotated views, whereas the true pose, mapped back through the known transform, is consistent across all of them. *Orbit voting* collects the tree's beam candidates from four rotated views and scores each candidate by its tree likelihood as seen from every view (Sec. 3.5). Finally, *orbit distillation* trains single-view students towards the orbit system's consensus (Sec. 3.6).

An audit on the synthetic-to-real benchmark SPEED+ [19] motivates the design (Sec. 4.3). Its target spacecraft is nearly symmetric. The real-domain error of the baseline is dominated by $180°$ flips about a few body axes, and an oracle that picks the best of the tree's own candidates reaches $9.8°$, against $26.9°$ for the prediction. The correct mode is usually present; it is not selected.

Our contributions are:

- A lightweight hierarchical 6D pose predictor on frozen DINOv3 and MoGe-2 features, with training-free per-level PoE fusion. In a matched comparison it is far more accurate than a flat 131k-anchor classifier, with 15× fewer parameters (Sec. 5).
- Symmetry-aware multi-path targets for hierarchical rotation trees. They reduce the SYMSOL-T error by 20–30% relative to single or canonical labels, and, unlike canonical labels, remain compatible with orbit voting (Sec. 5).
- Orbit voting with an exactly derived roll transform for hierarchical candidates, and orbit distillation into single-view students. On SPEED+ they repair 62–80% of severe flips while breaking 0.4% of good predictions, and reduce the real-domain error to 11.2°/11.1° (Sec. 4.3).
- An evaluation on four benchmarks (T-LESS, SYMSOL-T, YCB-V, SPEED+) with comparisons matched to published protocols where possible, three seeds, pre-registered criteria, selection on validation data only, and full reporting of negative results (Secs. 4 and 5).

Symmetry-aware losses, test-time augmentation and self-distillation are established ideas. Our claim concerns their exact geometric form for hierarchical rotation hypotheses and their measured effect on mode selection, not the ideas themselves. We do not claim a new state of the art: fine-tuned methods remain more accurate on every benchmark we use.

2. Related Work

Object pose estimation. Learning-based 6D pose estimators regress poses directly or predict 2D–3D correspondences followed by PnP [25]. The BOP benchmark [11] standardises evaluation on datasets such as T-LESS [10] and YCB-V [25], where textureless and symmetric objects remain the hardest cases. Synthetic-to-real transfer is studied in its purest form on SPEED+ [19], where models train on renders of a spacecraft and are tested on hardware-in-the-loop captures. There, keypoint-and-PnP pipelines with fine-tuned backbones lead [17, 18, 20, 23]; the most accurate recent result fine-tunes DINOv3 with LoRA and reports that a head on *frozen* DINOv3 features fails with frequent flips [3]. Direct regressors are faster but less accurate [1, 5, 26]. We keep the backbones frozen and ask how far selection-aware heads can take them.

Discrete and distributional rotation prediction. Uniform $SO(3)$ grids can be built from the Hopf fibration [27] or by recursive tesseract subdivision [15]; we adapt the latter to cell-centre representatives with unique paths. Grid- and implicit-density methods model multi-modal rotation posteriors, through MLP scores over grids [12, 16] or equivariant spherical features [14]; parametric alternatives include Bingham mixtures [4]. SpyroPose [7] evaluates a hierarchical $SE(3)$ pyramid coarse to fine, and LiePose [13] applies score-based diffusion on $SE(3)$ and handles symmetric objects well. Continuous rotation regression has known representation issues [28]. Riemannian [2] and discrete [6] flow matching provide generative alternatives; we tested both as refiners and report them as negative results.

Consistency, test-time augmentation and distillation. Averaging predictions over transformed inputs is a standard route to test-time robustness, and four-rotation test-time augmentation is also used for spacecraft keypoints [3]. Data distillation [21] turns multi-transform ensembles into training targets, a form of knowledge distillation [8]. Equivariant architectures build rotation consistency into the network [14]. Our orbit voting is a test-time augmentation over in-plane rotations, and orbit distillation is a data-distillation variant. What is specific here is the exact camera-roll transform for crops of off-axis objects, scoring with hierarchical tree likelihoods rather than averaging point estimates, and the target: a diagnosed mode-selection failure.

3. Method

Figure 1 gives an overview of training and inference.

3.1. Problem setup

Given a square crop of a known object around its bounding box (15% margin per side), we predict the rotation $q \in SO(3)$ (unit quaternion, $q \equiv -q$) of the object frame relative to the camera and the translation $t \in \mathbb{R}^3$. For objects that appear away from the optical axis (T-LESS, YCB-V), the tree predicts the *allocentric* rotation $R_{\text{ray}}(t)^\top R$, which removes the viewing-ray component so that the crop's appearance depends only on the predicted rotation; it is converted back exactly. Rotation error is $E_R = 2 \arccos |\langle \hat{q}, q \rangle|$, minimised over the object's symmetry set $\mathcal{S}$ when one exists ($\min_{S \in \mathcal{S}} E_R(\hat{q}, q\,S)$), and translation error is $E_T = \|\hat{t} - t\|_2$.

3.2. Tesseract tree on $SO(3)$

Following the recursive tesseract subdivision [15], a quaternion is assigned to a *chart* $c = \arg\max_i |q_i|$, sign-normalised so that $q_c > 0$ (this identifies q and $-q$), and projected to the cube $x = q/q_c$. Its three free coordinates in $[-1, 1]$ are halved recursively, so each level selects one of 8 children. We use *cell centres* (lifted back to the unit sphere) as representatives. Unlike corner representatives, every rotation then has a unique root-to-leaf path. With 4 charts and depth 5 there are $4 \cdot 8^5 = 131{,}072$ cells. The quantisation error of the leaf centre is 2.51° on average (3.97° p95, 6.20° worst case), and it halves per level. For objects without symmetry this floor bounds the fine accuracy of our ro-

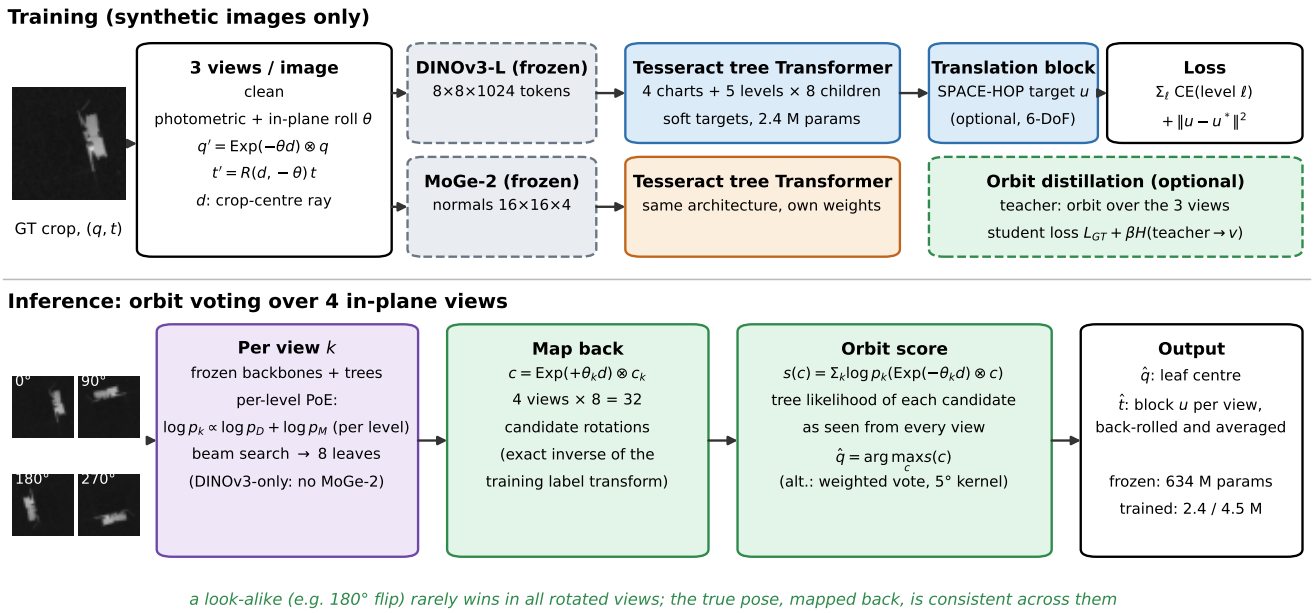


Figure 1. **Overview.** *Top:* training. Each synthetic crop is used as a clean view and as two augmented copies (photometric augmentation plus an in-plane roll θ with exactly transformed labels). Frozen DINOv3 and MoGe-2 features feed one tesseract tree Transformer each; an optional translation block reads the same tokens, and optional orbit distillation replaces the ground-truth target by an orbit teacher. *Bottom:* orbit voting at test time. Candidates from four rotated views are mapped back to a common frame and scored by their per-level PoE tree likelihood in every view.

tation results; for symmetric objects the nearest leaf to *any* equivalent rotation is closer.

Tree Transformer. For each backbone, a small decoder predicts the path autoregressively. The frozen tokens are layer-normalised, linearly projected to $d = 256$, given 2-D positional encodings and passed through one encoder layer. A causal decoder with 2 layers (4 heads, FFN 512) then reads a sequence that starts with a root token followed by one token per decided level. Each such token embeds the level and a 6-frequency Fourier encoding of the current cell centre, and cross-attends to the encoded image tokens. The root position predicts the chart (4 logits) and position ℓ predicts the child (8 logits). The model has 2.4 M parameters. We train with geodesic soft targets: at each level, the target over the candidate cells is $\propto \exp(-(\delta/\tau_\ell)^2)$, where δ is the angle between the ground truth and each cell centre. The widths τ_ℓ (39.3° for the root and level 1, 20.0°, 10.0°, 5.0°, 2.5°) halve with the cell size. Training uses teacher forcing on the ground-truth path and a cross-entropy at every level; inference is greedy or beam search, and the prediction is the leaf centre.

Frozen features. DINOv3 ViT-L/16 [22] last-layer patch tokens are average-pooled to an 8×8 grid (64×1024). For geometry, MoGe-2 [24] surface normals are masked by a foreground mask on which both backbones agree (the intersection of a DINOv3 foreground estimate and the MoGe-2 mask). They are then pooled to 16×16 cells, each holding three masked-normal channels and the mask fraction (256×4). This consensus mask removes most of MoGe-2's background leakage on real images (Appendix B). Features are extracted once; only the trees are trained.

Per-level product of experts. The two trees are fused without training. At every decoding step, the log-softmax distributions of both branches are added and renormalised: $\log p^{(\ell)} \propto \log p^{(\ell)}_{\text{DINO}} + \log p^{(\ell)}_{\text{MoGe}}$. Beam search then runs on the fused tree. We report this *PoE* variant and a faster *DINOv3-only* variant.

3.3. Symmetry-aware multi-path targets

For a symmetric object, the annotated rotation q is one arbitrary member of the equivalence set $\{q\,s : s \in \mathcal{S}\}$, where $\mathcal{S}$ holds the object-frame symmetries (continuous symmetries discretised). Training on q alone (*single label*) gives contradictory targets for identical images. The usual fix maps every label to a *canonical* member, *e.g.* the one closest to identity. This removes the contradiction but makes the target a discontinuous function of the image: two nearly identical views can receive canonical labels far apart, and the in-plane relation of Eq. (2) no longer holds between views.

Instead, we train the tree on all equivalent rotations. The loss is the soft-target cross-entropy of Sec. 3.2, averaged over the members of the set, each with teacher forcing along its own path:

$$\mathcal{L}_{\text{sym}} = \frac{1}{|\mathcal{S}'|} \sum_{s \in \mathcal{S}'} \mathcal{L}_{\text{GT}}(q\, s), \tag{1}$$

where $\mathcal{S}' = \mathcal{S}$ if $|\mathcal{S}| \le 8$ and otherwise 8 members are drawn per sample and step (an unbiased estimate). The tree learns to split its probability between the equivalent paths, which is the level-marginal target of the whole set. For objects without symmetry Eq. (1) reduces exactly to the standard loss. Right-multiplication by s commutes with the roll of Eq. (2), so augmented views and orbit voting are unchanged; we verified both identities numerically ($< 10^{-9}$).

3.4. Augmentation with exact in-plane labels

Each training image is used as a clean crop and as two pre-extracted augmented copies; one of the three is drawn at every step. The augmentations are photometric (brightness, contrast, gamma, blur, glare blobs, noise, random erasing, JPEG) plus an in-plane rotation of the crop by $\theta \sim \mathcal{U}(-180^\circ, 180^\circ)$ about its centre. Rotating the crop about its centre is equivalent to rotating the camera about the unit ray d through the crop centre, computed with the lens distortion removed. Points therefore transform as $p' = R(d, -\theta)\, p$, and the labels as

$$q' = \text{Exp}(-\theta\, d) \otimes q, \qquad t' = R(d, -\theta)\, t. \tag{2}$$

We derived this convention from the camera geometry rather than fitting it to a predictor. Projecting model keypoints with q' reproduces the warped image to 0.2° (p95 label error measured by PnP). For datasets whose objects lie far from the optical axis (T-LESS, YCB-V), an affine crop rotation is not exactly a camera roll (label error up to 6.7°). There we warp with the roll homography $KR(d, -\theta)K^{-1}$ instead, which makes Eq. (2) exact (verified to 0.000°).

3.5. Orbit voting

At test time we form $K{=}4$ views of the crop rotated by $\theta_k \in \{0^\circ, 90^\circ, 180^\circ, 270^\circ\}$. In view k the fused tree's beam search returns 8 leaves $c_{k,1..8}$, which are mapped back to the original frame by the inverse of Eq. (2), $c = \text{Exp}(+\theta_k d) \otimes c_k$, giving 32 candidates. Each candidate c is then scored by the tree likelihood of its own path when mapped into every view:

$$s(c) = \sum_{k=1}^{K} \log p_k\big(\text{path}(\text{Exp}(-\theta_k d) \otimes c)\big), \tag{3}$$

and predict $\hat{q} = \arg\max_c s(c)$. Here $\log p_k$ sums the fused root and child log-probabilities along the path, evaluated with teacher forcing. A 180° look-alike that wins in one view must also be likely in the other three to be selected, whereas the true pose is consistent by construction. We also evaluate a *weighted vote*, in which each candidate collects Gaussian-kernel support (5°) from the other candidates, weighted by their normalised beam probabilities within their view. Orbit voting has no parameters and multiplies inference cost by about $4\times$ (Tab. 5).

3.6. Orbit distillation

Orbit voting is more accurate than a single view, so we use it as a teacher. For every training image we run the frozen orbit system over its three available views (clean and two augmented, with known rolls). This gives 24 candidates in the clean frame with weights $w_j = \text{softmax}_j\, s(c_j)$. A new tree per branch is then trained from scratch with the same recipe and the loss

$$\mathcal{L} = \mathcal{L}_{\text{GT}}(v) - \frac{\beta}{L{+}1} \sum_j w_j \sum_{\ell=0}^{L} \log p^{(\ell)}\big(\text{pre}_\ell(\text{Exp}(-\theta_v d) \otimes c_j)\big), \tag{4}$$

with $\beta = 1$ and pre_ℓ the length-ℓ path prefix. Here v is the view drawn for the image, and the teacher candidates are transported into its frame. The student is a single-view model with unchanged inference cost; at test time it can be used alone or with orbit voting. This is a data-distillation scheme [21] applied to hierarchical rotation targets.

3.7. Translation block (6-DoF)

For translation we keep the SPACE-HOP parameterisation [5] unchanged. With box centre (b_x, b_y) and size (b_w, b_h), image size (W, H), intrinsics (f_x, f_y, c_x, c_y) and $\alpha = 1.6$, the target is $u = (U_x, U_y, U_z)$ with

$$U_x = \tfrac{1}{b_w}\big(f_x \tfrac{X}{Z} + c_x - b_x\big), \quad U_y = \tfrac{1}{b_h}\big(f_y \tfrac{Y}{Z} + c_y - b_y\big), \tag{5}$$

$$U_z = \tfrac{1}{2}\big(\tfrac{\alpha b_w}{W} + \tfrac{b_h}{H}\big) Z, \tag{6}$$

trained with $\|u - u^*\|^2$ and inverted in closed form. A translation block (one learned query cross-attending to the tree's encoded tokens, then an MLP) predicts u. It is trained either jointly with the tree ($\mathcal{L}_{\text{tree}} + \mathcal{L}_{\text{trans}}$) or on a frozen tree. Rolled views use t' from Eq. (2) with the unchanged box. With orbit voting, each view's $\hat{t}$ is rolled back and the four are averaged.

4. Experiments

4.1. Benchmarks and protocol

Benchmarks. We use four benchmarks that stress different kinds of ambiguity. **T-LESS** [10]: textureless industrial

parts with continuous, discrete or no symmetries. We use 7 objects spanning these classes (1, 4, 5, 11, 18, 21, 27), train on PBR renders and real training images, select on held-out PBR scenes, and test on all real BOP19 targets [11] of these objects. **SYMSOL-T**: five symmetric solids (tetrahedron, cube, icosahedron, cone, cylinder) with random translations, using the split of LiePose [13]. **YCB-V** [25]: household objects in cluttered real videos. We use 6 objects without declared symmetries (cracker box, mustard bottle, banana, mug, power drill, scissors), validate on held-out real sequences and test on the real keyframes. **SPEED+** [19]: a nearly symmetric spacecraft, 47,966 synthetic training and 11,994 synthetic validation images, tested on two real domains, *lightbox* (6,740) and *sunlamp* (2,791). Training uses synthetic images only.

Protocol. Experiments followed pre-registered protocols: question, arms, controls and pass criterion were fixed before each run, and protocol changes were recorded before the affected runs. Post-hoc analyses are labelled as such. We train 3 seeds and report mean $\pm$ seed standard deviation; a criterion passes if the difference exceeds the larger seed standard deviation. Checkpoints, hyper-parameters and variants are selected on validation data only; real test sets are never used for selection. One tree per object and branch is trained with the same recipe on every dataset. Errors are averaged per object and then over objects. Unless stated otherwise, crops come from ground-truth boxes (Sec. 6). Failed criteria and negative results are reported (Appendix C).

Implementation. Trees are trained for 100 epochs with batch 256, AdamW (learning rate 10^{-3}, weight decay 0.05, one-cycle schedule) and bf16, which takes 3–13 minutes per tree on one GPU. Timings are measured end-to-end at batch 1 on a shared NVIDIA RTX 6000 Ada and are upper bounds.

4.2. Comparison with published results

T-LESS (Tab. 1). Orbit voting raises recall at all thresholds for the single-label trees (R@10 68.1 $\rightarrow$ 73.5%), and canonical labels help the single view but break orbit voting (Sec. 5). LiePose is clearly more accurate, especially at fine thresholds (R@2 34.9% vs. 4.0%). Three factors are confounded in this gap: LiePose fine-tunes its backbone on all 30 objects, it masks the crop with the ground-truth visible mask, and our depth-5 leaves quantise rotations to 2.5° on average, which caps R@2.

SYMSOL-T (Tab. 2). Multi-path targets lower the error on every shape, from 2.62° to 2.07° on average. For symmetric shapes this is below the 2.31° error of the ground-truth rotation's own leaf, because the tree may pick the leaf closest to *any* equivalent rotation. LiePose remains about 4× more accurate; closing the gap needs a continuous refinement on top of the selected leaf.

YCB-V (Tab. 3). Trained on the real training videos alone, the trees fit their training frames but memorise video context and fail on new sequences (81°; Appendix G). Adding the standard synthetic renders, pasted onto real training-sequence backgrounds, reduces the test error to 10.9°, with 87% of predictions within 10°; five of the six objects reach

Method	Inference	R@2	R@5	R@10	mean (°)
LiePose [13]	1 sample	34.9	78.2	88.9	–
Ours, A0	1 view	2.3	27.0	68.1	17.9
	orbit (4)	4.0	36.0	73.5	16.6
Ours, A1	1 view	3.6	33.8	73.4	16.7
	orbit (4)	2.7	26.7	65.9	18.2
Ours, A2	1 view				
	orbit (4)				
Ours, A2 + distill.	1 view				
	orbit (4)				

Table 1. **T-LESS**, real BOP19 test targets of our 7 objects. Symmetry-aware rotation recall (%) at 2/5/10° and mean error, PoE variant, 3 seeds. LiePose numbers are its per-object results (Tab. 8 of [13]) averaged over the same 7 objects; it fine-tunes its backbone jointly on all 30 objects and multiplies the crop by the ground-truth visible mask. Ours: frozen backbones, ground-truth box, no mask. A0/A1/A2: single, canonical and multi-path targets (Sec. 3.3). Blank cells: runs not yet completed.

Method	tet	cube	icosa	cone	cyl	avg.
LiePose [13]	0.59	0.58	0.64	0.54	0.41	0.55
Ours, A0 + orbit	3.06	2.90	3.71	1.79	1.65	2.62
Ours, A2 + orbit	2.46	2.30	2.52	1.57	1.48	2.07
Ours, A2 + distill. + orbit						

Table 2. **SYMSOL-T** test, symmetry-aware mean rotation error (°) per shape. Ours: DINOv3 variant (selected on validation), orbit voting, 3 seeds. LiePose: SE(3) model, Tab. 3 of [13]. Blank cells: runs not yet completed.

Training data	Feat.	Inference	val	test	test acc@10
real videos	DINOv3	1 view	64.3 ±11.6	71.2 ±7.0	8%
	DINOv3	orbit (4)	63.0 ±2.4	69.4 ±5.2	9%
	PoE	1 view	73.5 ±6.8	81.1 ±7.3	16%
	PoE	orbit (4)	74.1 ±7.4	79.9 ±3.2	18%
+ synthetic (ours)	DINOv3	1 view	9.0 ±1.1	12.9 ±6.9	85%
	DINOv3	orbit (4)	11.5 ±1.8	13.4 ±4.7	83%
	PoE	1 view	10.5 ±0.9	10.9 ±4.1	87%
	PoE	orbit (4)	12.7 ±1.0	12.2 ±5.5	84%

Table 3. **YCB-V**, 6 objects, rotation error (°) on held-out real sequences (val) and real keyframes (test), and the share of test predictions within 10°. Adding synthetic renders composited on real training backgrounds turns a failure into a working model.

Method	Setting	E_R (°)		score	
		LB	SL	LB	SL
TangoUnchained* [20]	detector + keypoints/PnP, real images	3.19	4.30	0.073	0.090
lava1302* [20]	detector + keypoints/PnP, self-training	6.66	2.73	0.165	0.059
SPNv2 [17]	full image, no adaptation	6.48	11.07	0.142	0.230
SPNv2 + ODR* [17]	+ test-time adaptation	5.58	9.79	0.122	0.197
EagerNet [23]	detector + dense correspondences	1.75	2.66	0.039	0.059
SPNv3-B† [18]	GT crop, keypoints/PnP, fine-tuned	2.03	3.40	0.047	0.074
Church & Nikolian† [3]	GT crop, DINOv3-H+ LoRA, 12 passes	1.17	1.56	0.027	0.038
frozen DINOv3-L + head [3]	GT crop, frozen, 1 pass	15.47	24.49		
FastPose-ViT [1]	detector, direct regression	12.26	18.28		
PAID-ViT [26]	crop, direct regression	12.29	11.76	0.237	0.230
Ours, distilled + orbit	GT crop, frozen, 4 views	11.21	11.09		
Ours, 6-DoF (joint PoE) + orbit	GT crop, frozen, 4 views	12.11	11.73	0.241	0.238
Ours, distilled + orbit	synthetic-only detector, frozen				

Table 4. **SPEED+ real domains** (lightbox LB, sunlamp SL): mean rotation error and the official SPEED+ score (rotation error in radians plus normalised translation error, averaged over images; lower is better). Published values as reported by each source. * uses real or unlabelled test-domain images during training or adaptation. † ensemble of 3 models. Settings differ (detector or ground-truth crop, fine-tuned or frozen backbone), so only rows with the same setting are matched comparisons. Blank cells: runs not yet completed.

4–8°, and the thin, near-planar scissors remain the main failure (Tab. 17). Published YCB-V results use BOP or ADD(-S) metrics on all 21 objects with detected boxes, so we do not place our numbers in those tables.

SPEED+ (Tab. 4). With ground-truth crops, our frozen-feature system reaches 11.2°/11.1°, close to recent direct methods (PAID-ViT 12.3°/11.8°) and well ahead of a head on frozen DINOv3 features in the same crop setting (15.5°/24.5° [3]). Fine-tuned keypoint pipelines are far more accurate (1.2°–3.4°). Our official score with ground-truth crops (0.24) is similar to SPNv2 without adaptation on sunlamp and to PAID-ViT, and it is dominated by the rotation term.

4.3. Case study: a nearly symmetric target under domain shift

Diagnosis: real-domain errors are flips. Before changing the model, we audited the large real-domain errors of the clean (non-augmented) SPEED+ trees. Encoder and decoder round trips are exact, and the metric, rotation convention and data pipeline were ruled out. Chart seams were ruled out as well: flat Hopf-grid heads on the same features flip as often or more. Of the errors above 90°, 36–41% lie within 15° of 180°, against 20% for uniformly random wrong rotations. Their body-frame rotation axes cluster around a few axes of the near-symmetric spacecraft. Allowing a 180° rotation about one body axis lowers the lightbox mean of the DINOv3 tree from 38.9° to 23.5°. The correct mode is usually available: among the tree's beam candidates and their 180° flips, an oracle achieves 9.8° on lightbox, against 26.9° for the fused greedy prediction. Neither a continuous flow posterior, flow-based re-noise scoring, synthetic templates nor a discrete path flow selects it on real images (Appendix C). The problem is *mode selection* under domain shift.

Orbit voting and distillation. Table 5 summarises the progression. Augmentation alone (photometric plus roll, single view) cuts the real-domain error of the PoE system from 26.9°/36.7° to 17.3°/20.2°. Orbit voting then brings it to 12.3°/12.3° and the synthetic error to 3.87°. It repairs 62% and 80% of the severe flips on lightbox and sunlamp while breaking only 0.4% of good predictions (Tab. 13). The median errors on real images are 5.3° and 6.3°, so the remaining means are driven by a tail of failures. Orbit distillation improves every setting. The single-view students recover 13–43% of the orbit gain at single-view cost; for example, the PoE greedy error on lightbox drops from 17.3° to 15.9°. Students with orbit voting give the best system: 3.69° synthetic and 11.21°/11.09° on lightbox/sunlamp, about twice the seed std better than the undistilled orbit system, with fewer severe flips. For distillation, synthetic validation ranks the systems in the same order. For the orbit *scoring rule* it does not: the likelihood product was fixed in the base recipe before the scoring ablation was run, and synthetic validation would have selected the weighted vote instead (Sec. 5), which gives 3.27°/11.35°/11.91° with the distilled PoE students. MoGe-2 helps on real images: PoE with orbit voting reaches 12.3° on lightbox, against 17.4° for DINOv3-only. It costs about 50 ms per view, half of which is the CPU consensus-mask step. DINOv3-only orbit voting

Training	Feat.	Inference	Synth. val	Lightbox	Sunlamp	>150° (%)		ms
			mean E_R (°), 3 seeds			LB	SL	
Clean tree (no augmentation)	PoE	1 view	5.54 ±0.07	26.91 ±0.78	36.66 ±0.93	5.9	7.4	64
Augmented tree (base recipe)	DINOv3	1 view	4.84 ±0.05	22.55 ±0.73	22.00 ±0.70	6.0	3.2	15
Augmented tree (base recipe)	DINOv3	orbit (4)	4.21 ±0.04	17.40 ±0.68	15.27 ±0.64	4.8	2.0	70
+ orbit distillation	DINOv3	1 view	4.57 ±0.05	21.61 ±0.82	20.55 ±0.94	5.8	3.1	15
+ orbit distillation	DINOv3	orbit (4)	3.90 ±0.03	16.43 ±0.75	13.64 ±0.57	4.5	1.6	70
Augmented tree (base recipe)	PoE	1 view	4.85 ±0.03	17.33 ±0.31	20.15 ±0.57	4.2	3.3	64
Augmented tree (base recipe)	PoE	orbit (4)	3.87 ±0.03	12.30 ±0.50	12.25 ±0.37	3.0	1.7	279
+ orbit distillation	PoE	1 view	4.56 ±0.03	15.90 ±0.53	18.33 ±0.66	3.7	3.0	64
+ orbit distillation	PoE	orbit (4)	**3.69** ±0.02	**11.21** ±0.42	**11.09** ±0.49	2.6	1.4	279

Table 5. **SPEED+ rotation, ground-truth crops.** Mean E_R over 3 seeds (± seed std), fraction of severe flips (>150°), and end-to-end latency at batch 1. The clean tree is trained without augmentation. The DINOv3 student is distilled from a DINOv3-only orbit teacher, and the PoE students from the PoE orbit teacher. Inference cost is unchanged by distillation.

runs at 14 FPS and a single DINOv3 view at 68 FPS (Appendix A).

Full 6-DoF. Table 14 (supplementary) adds translation with the SPACE-HOP target [5]. Because our crops come from ground-truth boxes, the box alone already locates the target to 0.55 m. The translation block improves on this by about 7–10× on synthetic data (both pre-registered criteria pass). Joint training does not hurt rotation and even helps slightly on real images: PoE + orbit reaches 3.86°/12.11°/11.73°, against 3.87°/12.30°/12.25°. Orbit averaging lowers translation error by 10–25%. The DINOv3 block is the most accurate on synthetic data (0.059 m) and reaches 0.185/0.183 m on the real domains; the MoGe-2 block alone is weak for translation. An evaluation with a synthetic-only detector instead of ground-truth boxes is in progress (last row of Tab. 4).

5. Ablations

All ablations use the PoE variant, 3 seeds and the same evaluation run within each table.

Symmetry-aware targets. Table 6 compares the three training targets. Canonical labels (A1) remove the contradiction between equivalent labels and improve single-view accuracy on both datasets (SYMSOL-T 3.87° → 3.40°, T-LESS validation 12.3° → 11.5°). However, they break orbit voting: a canonical label is not related across in-plane rotations by Eq. (2), so the views disagree and the SYMSOL-T orbit error rises to 9.7°. Multi-path targets (A2) keep the relation and are the best single-view and orbit models on SYMSOL-T (2.72° and 2.10°; pre-registered criterion passes by about 100× the seed std).

		T-LESS (7 obj.)		SYMSOL-T (5)	
Training target	Inference	val	test	val	test
single annotated label	1 view	12.3 ±0.1	17.9 ±0.2	3.87 ±0.03	3.88 ±0.01
	orbit (4)	9.3 ±0.1	16.6 ±0.2	2.65 ±0.01	2.67 ±0.00
canonical label	1 view	11.5 ±0.0	16.7 ±0.1	3.40 ±0.01	3.39 ±0.01
	orbit (4)	10.7 ±0.0	18.2 ±0.7	9.73 ±0.05	9.69 ±0.05
all equivalents (ours)	1 view			2.72 ±0.01	2.71 ±0.01
	orbit (4)			2.10 ±0.01	2.11 ±0.01
+ orbit distillation	1 view				
	orbit (4)				

Table 6. **Training targets for symmetric objects** (symmetry-aware mean error, PoE). Single annotated label (A0), canonical label closest to identity (A1), and all equivalent rotations (A2, Eq. (1)); the last block adds orbit distillation on top of the selected target. Blank cells: runs not yet completed.

Head (params/branch)	Inference	Synth.	LB	SL
Tesseract tree (2.4 M)	1 view	4.85 ±0.03	17.33 ±0.31	20.15 ±0.57
Tesseract tree (2.4 M)	orbit	3.87 ±0.03	12.30 ±0.50	12.25 ±0.37
Flat Hopf, 131k (36.1 M)	1 view	8.95 ±0.06	20.56 ±0.70	27.16 ±1.04
Flat Hopf, 131k (36.1 M)	orbit	7.53 ±0.01	13.15 ±0.37	16.31 ±0.59

Table 7. **Tree vs. flat grid.** Same frozen features, augmentation, training budget and orbit inference. The flat head classifies 131,076 Hopf anchors, matching the tree's cell count.

Hierarchical tree vs. flat grid. We replace the tree with a count-matched flat classifier over 131,076 Hopf-fibration anchors [27], as in SPACE-HOP, keeping everything else identical (Tab. 7). The tree is far more precise (3.87° vs. 7.53° synthetic with orbit voting). It is better on real data in all comparisons and uses 15× fewer parameters. Orbit voting helps both heads, and the flat head with orbit voting even has slightly fewer severe flips on lightbox. Precision therefore comes from the tree, while flip repair comes from orbit voting.

Selection over 32 candidates	Synth.	LB	SL
1 view (greedy)	4.85 ±0.03	17.33 ±0.31	20.15 ±0.57
max-confidence	5.51 ±0.03	14.63 ±0.42	15.92 ±0.15
top-1 vote	4.37 ±0.03	13.24 ±0.42	13.38 ±0.57
weighted vote (32)	3.46 ±0.03	12.58 ±0.42	12.88 ±0.49
orbit product	3.87 ±0.03	12.30 ±0.50	12.25 ±0.37

Table 8. **Selection rule over the same 32 orbit candidates.**

Augmentation	1 view			orbit (4)		
	Syn.	LB	SL	Syn.	LB	SL
none	5.54 ±0.07	26.90	36.66	4.62 ±0.04	20.25	24.83
photometric	5.00 ±0.03	17.38	19.68	4.25 ±0.06	12.29	12.37
in-plane roll	4.91 ±0.00	24.30	31.52	3.78 ±0.03	18.69	20.70
both	4.85 ±0.03	17.33	20.15	3.87 ±0.03	12.30	12.25

Table 9. **Which augmentation matters.** *None* is the clean tree. The roll uses exact labels (Eq. (2)).

How to select among orbit candidates. With the same 32 candidates (Tab. 8), picking the most confident one is clearly worse than any form of agreement, and voting among the four views' top-1 predictions is better. Using all candidates helps most. The kernel-weighted vote is best on synthetic data (3.46°). The likelihood product of Eq. (3) is best on both real domains, but its advantage over the vote is within about one seed std. Its pre-registered criterion ("product beats voting on synthetic validation") *fails*, and under our selection protocol the weighted vote would be preferred. The product is reported in Tab. 5 only because it was fixed in the base recipe beforehand. We therefore claim multi-view consistency over hierarchical candidates, not a specific scoring rule.

Photometric vs. rotation augmentation. Table 9 separates the two augmentation families. Photometric augmentation with orbit voting reaches 12.3°/12.4° on real images, the same as the combined recipe. Rotation augmentation alone gives the best synthetic precision (3.78°) but leaves real errors at 18.7°/20.7°. Orbit voting on the clean tree already removes a quarter to a third of the real error (26.9° → 20.3°, 36.7° → 24.8°). The real-domain gain therefore comes from *lighting robustness plus orbit consistency*. The exact roll transform is needed for orbit voting itself; as a training augmentation it mainly improves synthetic precision. This corrects our initial hypothesis that rotation augmentation causes the flip repair.

Can training learn orbit consistency? On SPEED+ we compared paired views of the same image (a data control), a transported consistency loss on the leaf distribution (a standard equivariance regulariser), a *hierarchical* transported loss at every tree level, and orbit distillation (Tab. 15, supplementary). The hierarchical loss beats the base recipe and the paired-view control but not the leaf-only loss; its pre-registered criterion fails, so we make no hierarchical-transport claim. Both consistency losses mostly improve synthetic precision. Orbit distillation is the only objective that clearly improves the real domains (11.2°/11.1° with orbit voting).

Further ablations in the supplementary material cover the decoder lineage (MLP, GRU and Transformer decoders, hard vs. soft targets, beam search), fusion (single branches, learned fusion, weighted PoE), finer tokens, continuous flow posteriors and verifiers, discrete flow matching, oracle headroom, zero-shot transfer and timing.

6. Limitations

- **Ground-truth crops.** The reported results use ground-truth boxes; a synthetic-only detector for SPEED+ is being evaluated (Tab. 4). For translation in particular, the box alone gives 0.55 m on SPEED+, so a deployable claim needs detected boxes.
- **Translation beyond SPEED+.** The translation block is evaluated on SPEED+ only, so T-LESS and SYMSOL-T are compared on rotation.
- **Matched comparisons.** Published numbers use different crops, masks, object sets and fine-tuned backbones; Tabs. 1, 2 and 4 state each setting, and only rows with the same setting are matched.
- **Quantisation floor.** The depth-5 leaf quantisation (2.51° mean) bounds fine accuracy for asymmetric objects. Our best synthetic SPEED+ errors (3.3–3.7°) are close to it, and it caps fine-threshold recall on T-LESS. A continuous refinement on top of the selected mode is a natural next step; the refiners we tried helped only on synthetic data.
- **Per-object heads.** One tree is trained per object; there is no zero-shot transfer to new objects (Appendix H).
- **Cost.** Orbit voting costs 4× inference, and MoGe-2 fusion does not help on every dataset.
- **Timing.** Timings come from a shared GPU and are upper bounds.

7. Conclusion

A small tesseract tree on frozen DINOv3 and MoGe-2 features is a competitive basis for 6D object pose when its hypotheses are *selected* well. Three components target selection: multi-path targets that spread the training signal over all symmetric equivalents, orbit voting that scores hierarchical hypotheses for consistency across exactly mapped in-plane rotations, and orbit distillation that moves this consistency into single-view students. Multi-path targets improve every SYMSOL-T shape and, un-

like canonical labels, remain compatible with orbit voting. On the nearly symmetric SPEED+ target, orbit voting and distillation repair most real-domain flips. Synthetic training data turns YCB-V from a failure into a working model. Fine-tuned methods remain more accurate on all four benchmarks, and our evaluation is intentionally conservative; we hope the protocol, the matched comparisons and the negative results are as useful as the method.

References

[1] Pierre Ancey, Andrew Price, Saqib Javed, and Mathieu Salzmann. FastPose-ViT: A vision transformer for real-time spacecraft pose estimation. In *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 2026. 2, 6

[2] Ricky T. Q. Chen and Yaron Lipman. Flow matching on general geometries. In *International Conference on Learning Representations (ICLR)*, 2024. 2, 1

[3] John Church and Vazghen Nikolian. Vision foundation models with synthetic-only training for monocular spacecraft pose estimation, 2026. 2, 6

[4] Haowen Deng, Mai Bui, Nassir Navab, Leonidas Guibas, Slobodan Ilic, and Tolga Birdal. Deep Bingham networks: Dealing with uncertainty and ambiguity in pose estimation. *International Journal of Computer Vision*, pages 1–28, 2022. 2

[5] Arnabi Dutta, Arsh Abbas Naqvi, Kavinder Singh, Lavendra Gautam, Anil Singh Parihar, Shakti Bangare, and Ravi Kumar Lagisetty. SPACE-HOP: Spacecraft 6-DoF pose estimation using embedding-predictive pretraining and Hopf map. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, pages 10193–10201, 2026. 2, 4, 7

[6] Itai Gat, Tal Remez, Neta Shaul, Felix Kreuk, Ricky T. Q. Chen, Gabriel Synnaeve, Yossi Adi, and Yaron Lipman. Discrete flow matching. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024. 2, 1

[7] Rasmus Laurvig Haugaard, Frederik Hagelskjær, and Thorbjørn Mosekjær Iversen. SpyroPose: SE(3) pyramids for object pose distribution estimation. In *IEEE/CVF International Conference on Computer Vision Workshops (ICCVW)*, 2023. 1, 2

[8] Geoffrey Hinton, Oriol Vinyals, and Jeff Dean. Distilling the knowledge in a neural network, 2015. 2

[9] Geoffrey E. Hinton. Training products of experts by minimizing contrastive divergence. *Neural Computation*, 14(8): 1771–1800, 2002. 1

[10] Tomáš Hodaň, Pavel Haluza, Štěpán Obdržálek, Jiří Matas, Manolis Lourakis, and Xenophon Zabulis. T-LESS: An RGB-D dataset for 6D pose estimation of texture-less objects. In *IEEE Winter Conference on Applications of Computer Vision (WACV)*, pages 880–888, 2017. 2, 4

[11] Tomáš Hodaň, Martin Sundermeyer, Bertram Drost, Yann Labbé, Eric Brachmann, Frank Michel, Carsten Rother, and Jiří Matas. BOP challenge 2020 on 6D object localization. In *European Conference on Computer Vision Workshops (ECCVW)*, pages 577–594, 2020. 2, 5

[12] Timon Höfer, Benjamin Kiefer, Martin Messmer, and Andreas Zell. HyperPosePDF: Hypernetworks predicting the probability distribution on SO(3). In *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, pages 2369–2379, 2023. 2

[13] Tsu-Ching Hsiao, Hao-Wei Chen, Hsuan-Kung Yang, and Chun-Yi Lee. Confronting ambiguity in 6D object pose estimation via score-based diffusion on SE(3). In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024. 1, 2, 5

[14] David M. Klee, Ondrej Biza, Robert Platt, and Robin Walters. Image to sphere: Learning equivariant features for efficient pose prediction. In *International Conference on Learning Representations (ICLR)*, 2023. 2

[15] Gerhard Kurz, Florian Pfaff, and Uwe D. Hanebeck. Discretization of SO(3) using recursive tesseract subdivision. In *IEEE International Conference on Multisensor Fusion and Integration for Intelligent Systems (MFI)*, 2017. 1, 2

[16] Kieran Murphy, Carlos Esteves, Varun Jampani, Srikumar Ramalingam, and Ameesh Makadia. Implicit-PDF: Nonparametric representation of probability distributions on the rotation manifold. In *International Conference on Machine Learning (ICML)*, 2021. 1, 2

[17] Tae Ha Park and Simone D'Amico. Robust multi-task learning and online refinement for spacecraft pose estimation across domain gap. *Advances in Space Research*, 73(11): 5726–5740, 2024. 2, 6

[18] Tae Ha Park and Simone D'Amico. Bridging the domain gap for flight-ready spaceborne vision. *Journal of Spacecraft and Rockets*, 63(1):289–307, 2026. 2, 6

[19] Tae Ha Park, Marcus Märtens, Gurvan Lecuyer, Dario Izzo, and Simone D'Amico. SPEED+: Next-generation dataset for spacecraft pose estimation across domain gap. In *IEEE Aerospace Conference (AERO)*, pages 1–15, 2022. 1, 2, 5

[20] Tae Ha Park, Marcus Märtens, Mohsi Jawaid, et al. Satellite pose estimation competition 2021: Results and analyses. *Acta Astronautica*, 204:640–665, 2023. 2, 6

[21] Ilija Radosavovic, Piotr Dollár, Ross Girshick, Georgia Gkioxari, and Kaiming He. Data distillation: Towards omni-supervised learning. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2018. 2, 4

[22] Oriane Siméoni et al. DINOv3, 2025. 1, 3

[23] Maximilian Ulmer, Maximilian Durner, Martin Sundermeyer, Manuel Stoiber, and Rudolph Triebel. 6D object pose estimation from approximate 3D models for orbital robotics. In *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 10749–10756, 2023. 2, 6

[24] Ruicheng Wang et al. MoGe-2: Accurate monocular geometry with metric scale and sharp details. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2025. 1, 3

[25] Yu Xiang, Tanner Schmidt, Venkatraman Narayanan, and Dieter Fox. PoseCNN: A convolutional neural network for 6D object pose estimation in cluttered scenes. In *Robotics: Science and Systems (RSS)*, 2018. 2, 5

[26] Zongwu Xie, Yifan Yang, Yonglong Zhang, Guanghu Xie, Yang Liu, and Shuo Zhang. Precision-aware illumination-disentangled vision transformer for spacecraft 6D pose estimation, 2026. 2, 6

[27] Anna Yershova, Swati Jain, Steven M. LaValle, and Julie C. Mitchell. Generating uniform incremental grids on SO(3) using the Hopf fibration. *The International Journal of Robotics Research*, 29(7):801–812, 2010. 2, 7

[28] Yi Zhou, Connelly Barnes, Jingwan Lu, Jimei Yang, and Hao Li. On the continuity of rotation representations in neural networks. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2019. 2

Selecting the Right Mode: Symmetry-Aware, Orbit-Consistent Tesseract Trees on Frozen Foundation Features for 6D Object Pose Estimation

Supplementary Material

A. Timing

System	batch 1		batch 64	
	ms	FPS	ms/img	img/s
DINOv3, 1 view	14.7	68.1	2.9	347
DINOv3 + orbit	69.6	14.4	12.1	82
PoE, 1 view	64.0	15.6	50.0	20
PoE + orbit	278.5	3.6	206.8	5

Table 10. **End-to-end latency** of the augmented trees (shared RTX 6000 Ada; upper bounds; image already decoded).

At batch 1 (Tab. 10), a DINOv3 forward pass takes 6.4 ms and one tree 7.4 ms. MoGe-2 adds 21 ms for the network and 20 ms for the CPU consensus-mask step per view, which is the main optimisation target. Orbit voting multiplies the cost by 4.4–4.7×. Batched evaluation reduces tree cost to 0.1–0.3 ms per image.

B. Model lineage and fusion

Design (clean trees, PoE greedy)	Synth.	LB	SL
MLP path decoder, hard targets	6.24	31.1	43.7
GRU path decoder	6.15	–	40.6
MLP decoder, geodesic soft targets	6.07	28.2	42.1
Transformer decoder, hard targets	5.69	31.6	41.8
Transformer + soft targets	**5.54**	**26.9**	**36.7**
+ beam 8 instead of greedy	no gain	–	–
+ tangent residual at the leaf	−0.13	±0.0	±0.0
16×16 instead of 8×8 DINOv3 tokens	5.56	27.0	36.0
DINOv3 tree alone	5.35	37.1	–
MoGe-2 tree alone	–	43.7	–
Learned concat fusion, one Transformer (8×8)	5.34	40.8	–
Equal per-level PoE (same grid)	5.58	27.9	–
Weighted PoE, w_{DINO}=0.8 (proxy-selected)	5.14	32.0	36.7

Table 11. **Design steps** on clean (non-augmented) trees, seed means. Dashes: not evaluated in that study.

Table 11 lists the design steps that led to the base tree. The Transformer decoder with geodesic soft targets was best; beam search at full data brought no gain, and a tangent residual only refined errors within the correct cell. Fusion is strongly asymmetric across domains. A learned fusion that feeds concatenated tokens into one Transformer matches DINOv3 alone on synthetic data but is much worse on real images (40.8° vs. 27.9° for PoE on the same grid). We interpret this as the learned fusion over-fitting render statistics, while training-free PoE keeps the branches' errors partly independent. A weighted PoE selected on the rotated validation proxy (w=0.8) is 5° worse on lightbox than equal weights. Following our protocol, we did not use real data to choose another weight. The consensus foreground mask reduces MoGe-2's background leakage on real images from 0.52 to 0.12 (lightbox) and from 0.57 to 0.12 (sunlamp). It is what makes the MoGe-2 branch useful.

C. Negative results

Approach (on clean trees unless noted)	Synth.	LB	SL
Clean PoE tree (reference)	5.54	26.9	36.7
$SO(3)$ flow posterior on tree memory, KDE mode	5.22	29.8	37.6
best of 32 samples (*oracle*)	2.60	9.7	10.6
Flow re-noise verifier over beams + flips + corners	4.78	26.5	32.7
best candidate (*oracle*)	2.25	7.9	11.4
Kurz corner tree [15]	5.20	28.8	39.1
Tree distilled from flow masses (β=1, 1 seed)	5.62	26.2	37.7
Discrete flow matching on tree paths	7.22	29.8	39.8
Aug. DINOv3 tree (reference)	4.84	22.6	22.1
+ $SO(3)$ flow, KDE	4.10	25.2	24.4

Table 12. **Refiners and verifiers that did not fix real flips** (seed means). Oracles are not deployable and show headroom only.

Table 12 collects approaches that improved synthetic accuracy but not, or only slightly, real accuracy. A Riemannian flow-matching posterior [2] conditioned on the tree's encoder memory contains good samples: its best-of-32 oracle reaches 9.7° on lightbox. It also assigns the ground truth a much higher log-likelihood than the tree does (3.85 vs. 0.98 nats on lightbox), and its 90% regions over-cover (0.98). Yet its mode estimate is worse than the tree's. Re-scoring an expanded candidate set (beam, 180° flips, cell corners) by re-noise/return consistency helps synthetic data and sunlamp, but does not repair lightbox flips. Synthetic render-and-compare templates also failed to choose the right flip. A discrete flow on the tesseract path [6] was clearly worse. We note that interpolating path tokens is not geometry-preserving: 45–53% of intermediate paths lie more than 30° from both endpoints. Distilling flow masses into the tree did not pass its criterion. An oracle that stops decoding at the best depth would gain only 0.51° on synthetic data, so adaptive depth was not pursued. These results support the diagnosis of Sec. 4.3: the right mode is reachable, but scorers trained on clean synthetic features cannot select it on real images. Lighting-robust training and multi-

view consistency can.

D. Flip transitions on SPEED+

System	Lightbox		Sunlamp	
	fixed	broken	fixed	broken
Aug. tree, 1 view	51% of 401	1.0%	59% of 206	1.0%
Aug. tree + orbit	62% of 401	0.4%	80% of 206	0.4%
Aug. DINOv3 + orbit	52% of 401	1.3%	74% of 206	0.5%

Table 13. **Flip transitions relative to the clean PoE tree.** *Fixed*: share of the clean tree's $>150^\circ$ errors that become $<20^\circ$ (denominator given). *Broken*: share of its $<20^\circ$ predictions that become $>150^\circ$.

E. Orbit distillation and 6-DoF details

Arm	Feat.	$\hat{t}$	E_T (m)			E_R (°), orbit		
			Syn.	LB	SL	Syn.	LB	SL
Box only	–	–	0.561	0.547	0.553	–	–	–
Frozen	DINOv3	1 view	0.090	0.233	0.240	4.21	17.40	15.27
		orbit	0.068	0.205	0.205			
Frozen	PoE	1 view	0.108	0.206	0.232	3.87	12.30	12.25
		orbit	0.089	0.184	0.206			
Joint	DINOv3	1 view	0.078	0.209	0.212	4.19	17.31	14.87
		orbit	**0.059**	0.185	**0.183**			
Joint	PoE	1 view	0.111	0.196	0.221	3.86	12.11	11.73
		orbit	0.093	**0.175**	0.194			

Table 14. **SPEED+ 6-DoF** with the SPACE-HOP translation target, ground-truth crops. Rotation uses PoE or DINOv3 orbit voting. Translation ($\hat{t}$) is predicted from one view or as the orbit average; seed std of $E_T \leq 0.005$ m. *Box only* uses no image: the object centre is placed at the box centre and the depth at the training mean of U_z.

Training objective	1 view, DINOv3		PoE + orbit		
	Syn.	Syn.-rot.	Syn.	LB	SL
Base recipe (GT loss)	4.84 ±0.05	9.88 ±0.22	3.87	12.30	12.25
+ paired views	4.96 ±0.06	10.04 ±0.21	3.94	12.38	12.45
+ leaf-level transported consistency	4.59 ±0.04	9.43 ±0.14	3.71	12.41	11.74
+ hierarchical transported consistency	4.65 ±0.03	9.57 ±0.06	3.74	12.49	11.73
+ orbit distillation	4.57 ±0.05	9.33 ±0.18	**3.69**	**11.21**	**11.09**

Table 15. **Training-time consistency.** Single-view DINOv3 trees on synthetic and rotated synthetic validation (left) and PoE + orbit (right). The consistency losses transport each view's beam distribution into another view's frame via Eq. (2).

Retention of the orbit gain by the single-view students, defined as (base greedy − student greedy)/(base greedy − base orbit), is 43%/18%/22% (DINOv3) and 30%/28%/23% (PoE) on synthetic/lightbox/sunlamp, and 13–15% on the rotated proxy. For the joint 6-DoF model, the ESA-style score (rotation error in radians plus normalised translation error, with the challenge precision floors) is 0.084/0.241/0.238 for PoE.

F. Per-object results

Object	PoE+orbit	DINOv3+orbit
T-LESS		
1	23.82	25.00
4	14.50	15.69
5	6.22	7.07
11	11.90	12.92
18	14.38	14.78
21	34.78	37.24
27	10.36	12.43
SYMSOL-T		
cone	1.83	1.79
cube	2.72	2.90
cyl	1.67	1.65
icosa	3.81	3.71
tet	3.35	3.06
YCB-V		
003_cracker_box	85.27	74.61
006_mustard_bottle	97.96	87.29
011_banana	61.63	83.23
025_mug	24.25	22.76
035_power_drill	41.34	25.07
037_scissors	169.07	123.72

Table 16. **Per-object test error** of the single-label (A0) trees (3-seed mean; symmetry-aware for T-LESS and SYMSOL-T; YCB-V trained on real videos only).

On T-LESS (Tab. 16), most of the test error comes from object 21 (no symmetry metadata, near-symmetric; 18% of predictions above 90°) and object 1 (continuous symmetry). Continuous symmetries are evaluated on a 10°-step discretisation of the symmetry set, which adds up to 5° of artificial error for objects 1 and 4 and for the cone and cylinder.

G. YCB-V diagnosis

On YCB-V we checked four things:

1. The trees fit their training frames: 3.8° and 3.9° on mustard bottle and mug.
2. Test poses are covered by training: the median nearest training rotation is about 5° away for cracker box, mustard bottle and banana, whose test errors are still 75–87°.
3. Every crop contains all of the object's label pixels.
4. The ground-truth rotations agree with the segmentation: 65–98% of the projected model points fall on the object mask, against 28–83% for a transposed-rotation control.

The failure is therefore memorisation of the 16–20 cluttered training videos per object. The object covers only 5–40% of the crop and the DINOv3 tokens are not masked. This

is not a label or pipeline error. Standard YCB-V training adds the synthetic `data_syn` renders, whose backgrounds are black. We pasted each render onto a real frame drawn only from the training sequences (never validation or test), capped at 12,000 crops per object as for the real data, and retrained with the unchanged recipe. Table 17 shows the effect per object.

	real videos		+ synthetic	
Object	DINOv3	PoE	DINOv3	PoE
003_cracker_box	77.7	78.7	5.1	5.7
006_mustard_bottle	101.5	102.4	6.1	7.6
011_banana	79.9	77.6	6.1	5.3
025_mug	24.6	28.4	6.5	5.6
035_power_drill	30.3	38.2	4.3	4.4
037_scissors	113.2	161.5	49.4	36.9

Table 17. **YCB-V per-object test error** (°, single view, 3 seeds) without and with synthetic training data.

H. Zero-shot transfer of the frozen SPEED+ trees

Dataset	Raw E_R		Frame-free pair error		
	frozen	chance	frozen	chance	constant
T-LESS	117.5	103.6	95.4 $\pm$3.0	104.5	88.7
SYMSOL-T	49.4	53.7	47.8 $\pm$0.5	53.7	53.8
YCB-V	115.6	125.8	101.6 $\pm$2.8	125.8	57.8

Table 18. **Zero-shot**: frozen SPEED+ PoE trees with orbit voting, no retraining, test split. The pair error $\min_S \angle(\hat{R}_i \hat{R}_j^\top, R_i S R_j^\top)$ over random same-object pairs cancels the unknown offset between object frames. *Chance*: uniformly random predictions. *Constant*: one rotation predicted for every image.

Object frames are arbitrary, so raw errors are expected near chance. The frame-free pair error (Tab. 18) is therefore the meaningful measure. On T-LESS and YCB-V it is *worse* than predicting one constant rotation, because both datasets have strong pose priors. On SYMSOL-T, where rotations are uniform, the trees beat chance only because of one shape: the cube (27.8° vs. 41.2°), plausibly because the spacecraft body is box-like. There is no general zero-shot transfer: the trees must be retrained per object.

I. Disclosures and reproducibility

- **Late feature-equivalence check.** The augmented and rotated views are extracted by a compact feature extractor. A pre-registered check against the full feature caches was run after the main results. It failed its strict element-wise tolerance: the maximum standardised difference was 0.51 for MoGe-2 features (p99 0.004; correlation 0.999998). Its functional follow-up showed that no PoE prediction changed on 512 crops, and that single-branch greedy leaves changed for 1/512 (DINOv3) and 3/512 (MoGe-2) crops.
- **SYMSOL-T bug.** Nine MoGe-2 SYMSOL-T trees first trained to NaN: always-empty border cells had a standard deviation near 10^{-6}, and float16 overflowed. They were retrained with a standard-deviation floor of 10^{-2}; all other trees were unaffected. Only the corrected evaluation is reported.
- **YCB-V paths.** A file-path error in the YCB-V manifest (colour images are JPEG) was fixed before any YCB-V result.
- **Post-hoc control.** The constant-prediction control in Tab. 18 was added after the T-LESS zero-shot result and is labelled as post hoc.

Every number in the results tables is generated by a script directly from the evaluation logs of the named runs.